

Essential Summary

A Modified Generalized Harmonic Function Perturbation Method and Its Application in Analyzing Generalized Duffing–Harmonic –Rayleigh–Liénard Oscillator

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1 Background and Motivation

This is the **foundational paper** of the MGHFP method series. Many classical analytical methods can analyze nonlinear oscillators quantitatively — but only when the restoring force and damping are simple. Once the oscillator becomes complicated, these methods **cannot execute their procedures symbolically** without first assigning numerical values to all system parameters. They can only produce isolated solution snapshots, never a continuous analytical picture of how solutions evolve with changing parameters.

This work introduces the **modified generalized harmonic function perturbation (MGHFP) method** by embedding the composite Simpson quadrature formula into the classical GHFP framework, enabling the entire perturbation procedure to proceed purely symbolically — no parameter pre-assignment required. This is the foundation upon which all subsequent MGHFP applications (IJNLM 2025, Phys. Scr. 2026) are built.

2 System Model

The generalized Duffing–Harmonic–Rayleigh–Liénard (DHRL) oscillator (Eq. 1 from the original paper):

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = \varepsilon(\mu_c + \mu_2 x^2 + \mu_4 x^4 + \mu_6 x^6 + \mu_{22} \dot{x}^2 + \mu_{24} \dot{x}^4) \dot{x} \quad (1)$$

where λ, μ, ν are stiffness parameters, and the right side contains a **6-term generalized nonlinear damping** combining Rayleigh terms ($\dot{x}^2, \dot{x}^4$) and Liénard terms (x^2, x^4, x^6). The rational restoring force makes this oscillator a maximally challenging test case.

3 The MGHFP Method

3.1 How It Works

The classical GHFP method proceeds via: (1) nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$, (2) Fourier expansion of the solution $x = a \cos^2 \Phi + b$, (3) perturbation expansion $a = a_0 + \varepsilon a_1 + \dots$, and (4) evaluation of integrals to obtain Fourier coefficients. Step (4) is the bottleneck — for complicated oscillators, these integrals **cannot be evaluated analytically**.

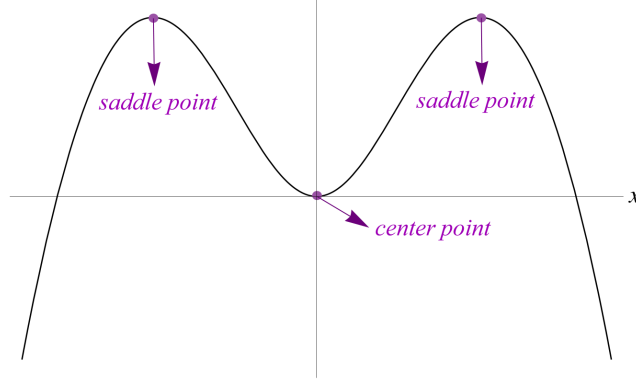


Figure 1: **Figure 6.** Potential portrait of the DHRL oscillator. The rich potential landscape enables homoclinic and heteroclinic bifurcations, making this an ideal testbed for the MGHFP method.

3.2 The MGHFP Fix — Composite Simpson Integration

The MGHFP method discretizes the integrals using the composite Simpson quadrature formula:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}} f(x_i) + f(x_n) \right] \quad (2)$$

By dividing the integration interval into eight equal subintervals, the Fourier coefficients p_{2i} become **explicit algebraic functions** of a_0 , b , and system parameters. The entire perturbation procedure now runs purely symbolically.

3.3 Nonlinear Time Transformation Details

The solution is assumed as $x(t) = a \cos^2 \varphi(t) + b$, with the nonlinear time transformation:

$$\frac{d\varphi}{dt} = \Phi(\varphi) = \Phi_0 + \varepsilon \Phi_1 + \dots \quad (3)$$

The zero-order frequency function is:

$$\Phi_0(\varphi) = \frac{\sqrt{2[V(a_0 + b) - V(a_0 \cos^2 \varphi + b)]}}{x'_0(\varphi)} \quad (4)$$

where $V(x) = \int g(x) dx$ is the potential energy. For complicated $g(x)$, this expression is analytically intractable — which is precisely why the Simpson discretization is needed. After discretization, Φ_0 is expanded as a Fourier series $\Phi_0(\varphi) = \sum_{i=0}^M (p_{2i} \cos 2i\varphi + q_{2i} \sin 2i\varphi)$ with M chosen appropriately (here $M = 4$).

4 Key Equations

4.1 Amplitude–Parameter Relation (Central Result)

The analytical relationship between limit cycle amplitude A and control parameter μ_c :

$$\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)} \left(E_2\mu_2 + E_4\mu_4 + E_6\mu_6 + E_{22}\mu_{22} + E_{24}\mu_{24} \right) \quad (5)$$

where E_i are functions of a_0 , b , and the Fourier coefficients p_{2i} computed via composite Simpson integration. Given system parameters, the limit cycle amplitude is obtained **analytically** — no numerical integration needed.

4.2 Stability Criterion

$$h_0 = \frac{1}{\pi} \int_0^\pi \left[\mu_c + \sum \mu_i (a_0 \cos^2 \varphi + b)^i + \sum \mu_{ij} (a_0 \cos^2 \varphi + b)^i (\dot{x}_0)^2 \right] d\varphi \quad (6)$$

$h_0 > 0$: unstable; $h_0 = 0$: semi-stable (bifurcation point); $h_0 < 0$: stable.

4.3 Homoclinic and Heteroclinic Bifurcation Thresholds

Setting the limit cycle amplitude equal to the saddle point coordinate in Eq. (5) yields the critical μ_c^{hom} for homoclinic bifurcation. For heteroclinic bifurcation, the amplitude is set to the coordinates of both connected saddle points, yielding critical μ_c^{hetero} values. The variation of μ_c^{hetero} with respect to system parameters is analyzed in Figure 8.

5 Example 1: Limit Cycle Evolution (Section 4.2)

The following parameters are used: $\lambda = 2$, $\mu = 1$, $\nu = 5$, $\mu_2 = -0.88$, $\mu_4 = 0.39$, $\mu_6 = -0.1$, $\mu_{22} = 0.17$, $\mu_{24} = 0.18$.

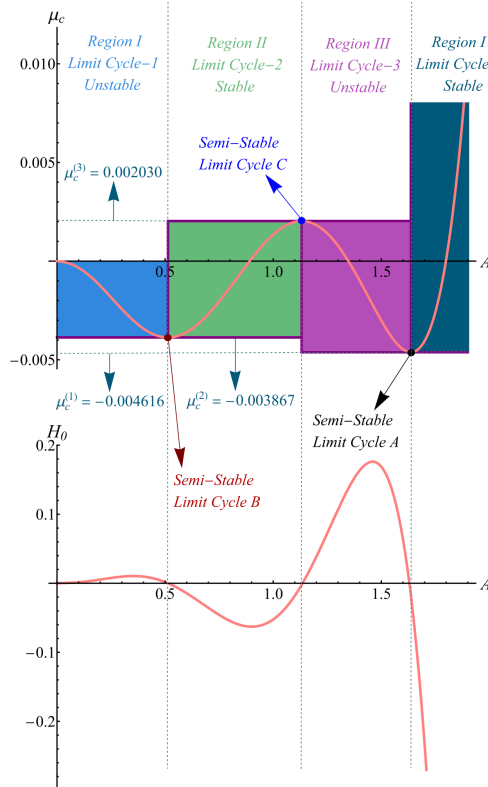


Figure 2: **Figure 2.** Evolution of limit cycles in the parametric plane (μ_c - A curve). The complete lifecycle is visible: four regions (I–IV) are identified based on the number and stability of limit cycles. Region I: no limit cycle; Region II: one stable cycle; Region III: two cycles (stable + unstable); Region IV: one stable cycle with different amplitude range. Semi-stable bifurcation points A and B mark the transitions between regions.

Figure 2 reveals the complete limit cycle lifecycle in parametric space:

- **Region I** ($\mu_c < \mu_c^A$): no limit cycle exists.
- **At Point A**: a **semi-stable** limit cycle emerges (outer-stable, inner-unstable).

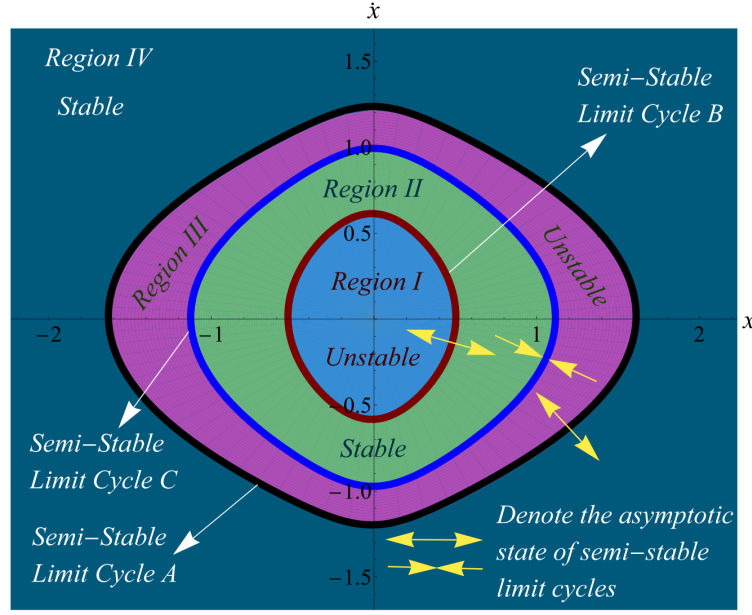


Figure 3: **Figure 3.** Phase portraits corresponding to the four regions in Figure 2. Each subfigure shows the limit cycles in phase space for a representative μ_c value within that region. The method accurately captures the transition from no limit cycle to stable, to coexisting stable+unstable, and back to stable-only behavior.

- **Region II** ($\mu_c^A < \mu_c < \mu_c^B$): two coexisting limit cycles — a smaller unstable cycle and a larger stable cycle.
- **At Point B**: a second semi-stable bifurcation occurs.
- **Region III** ($\mu_c^B < \mu_c < 0$): one stable and one unstable limit cycle coexist.
- **Region IV** ($\mu_c > 0$): only a single stable limit cycle persists.

Figure 3 provides the corresponding phase portraits in each region, visually confirming the parametric analysis.

6 Example 2: Heteroclinic Bifurcation (Section 4.3)

The heteroclinic bifurcation analysis focuses on oscillator (68) from the paper, which involves a different parameter configuration. The key innovation here is the ability to derive **analytical expressions** for the heteroclinic bifurcation parameter μ_c^{hetero} , enabling quantitative study of its variation with respect to system parameters.

7 Key Findings

1. **First purely symbolic GHFP variant:** The introduction of composite Simpson quadrature into the GHFP framework makes the proposed method the first variant capable of **purely symbolic execution** for complicated oscillators — no parameter pre-assignment needed. This is the foundational innovation that enables all subsequent MGHFP work.
2. **Complete lifecycle from parametric plane (Figures 2–3):** The μ_c – A curve reveals the entire lifecycle of each limit cycle, with four distinct regions separated by semi-stable bifurcation points. Phase portraits in each region visually confirm the parametric analysis.

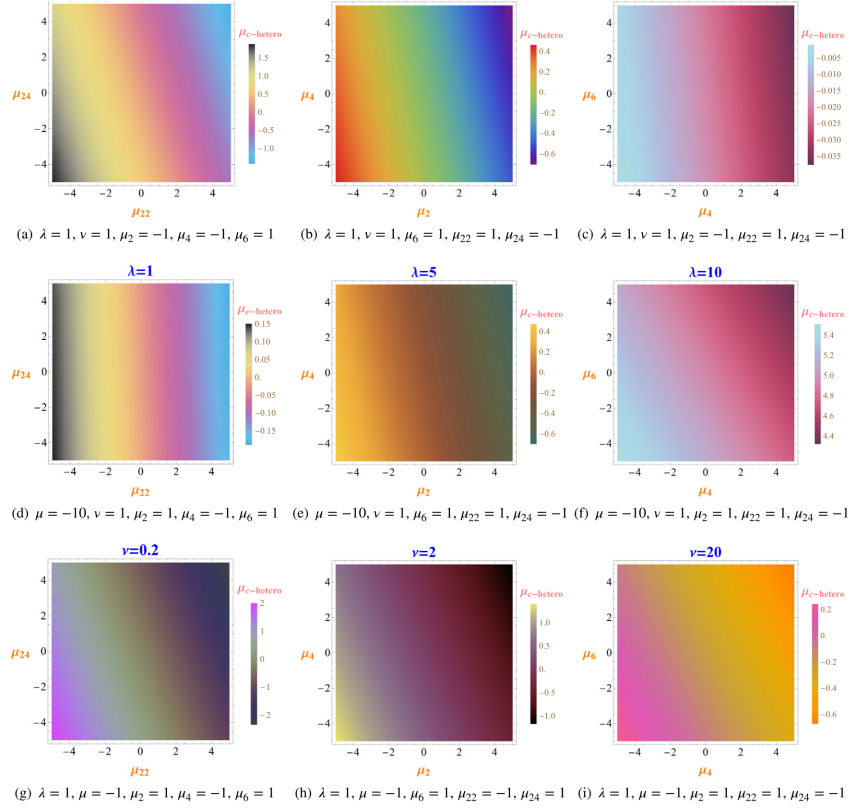


Figure 4: **Figure 8.** Variation of the heteroclinic bifurcation parameter μ_c^{hetero} with system parameters in different conditions. Each subfigure shows how μ_c^{hetero} changes as one damping coefficient is varied while others are fixed. The MGHFP method enables this multi-parameter analysis analytically — a capability classical GHFP lacks.

3. **Homoclinic and heteroclinic bifurcations predicted (Figures 8–9):** Both types of global bifurcation are quantitatively analyzed. The analytical expression for μ_c^{hetero} enables systematic study of its dependence on all damping coefficients simultaneously — a multi-parameter capability unique to MGHFP.
4. **Method validated on maximally challenging test case:** The DHRL oscillator combines rational restoring force $\frac{\lambda x + \mu x^3}{1 + \nu x^2}$ with six-term generalized Rayleigh–Liénard damping. All analytical predictions agree with fourth-order Runge–Kutta results.
5. **Foundation for the MGHFP series:** This paper establishes the framework subsequently extended to SD oscillators with quartic damping (IJNLM 2025, with Padé enhancement) and coupled SD oscillators with two irrational terms (Phys. Scr. 2026).

8 Engineering Significance

The generalized DHRL oscillator models realistic scenarios in MEMS resonators (squeeze-film damping), vibration isolators (nonlinear electromagnetic damping), and energy harvesting (self-excited systems). The MGHFP method provides the first analytical tool capable of producing continuous amplitude–parameter curves for such systems — a capability previously restricted to qualitative methods or case-by-case numerical studies.

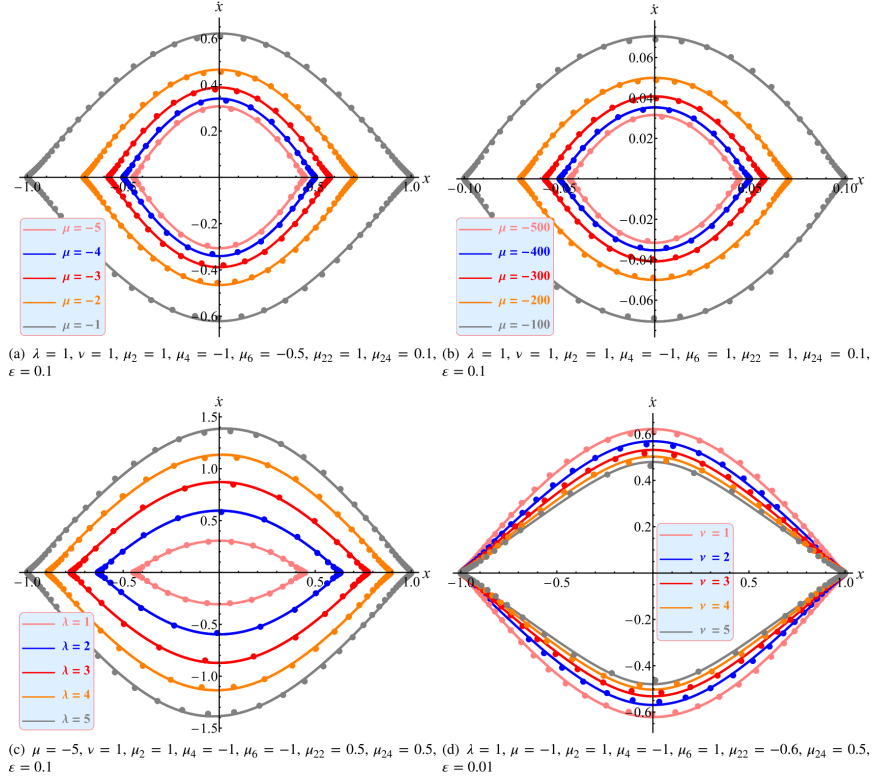


Figure 5: **Figure 9.** Phase portraits of heteroclinic solutions with different parameters. Each subfigure compares the analytical MGHFP solution with the Runge–Kutta numerical reference. The excellent agreement validates the method’s accuracy for heteroclinic orbit prediction.

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